

The 3D Rage Pro released in 1997 improved, offering a fill rate equal to that of the original 3dfx Voodoo Graphics, and a proper 1.2 M triangle/s hardware setup engine. The Rage Pro was held back by poor drivers. It was only in 1999, almost two years after the original launch, that the drivers finally came good, delivering a 20 to 40 per cent gain over the originals. ATi then learnt to prioritise driver development.

Released in 2000, the Radeon core became known in later versions as the 7000 and 7500, reflecting its DirectX 7-compliant features set. The core established a number of notable firsts, such as a complete DX7 bump mapping implementation, hardware 3D shadows, hardware per-pixel video-de-interlacing, and a reasonably good implementation of many advanced DX8 pixel shader effects.

nVidia products of that period delivered greater raw power in terms of fill rates, but ATi started to open up a clear quality and shader performance advantage. Still, it was only the three series of Detonator drivers that unlocked an unexpected 20 per cent performance gain that enabled nVidia to hold on to the performance crown.

ATi proved the Radeon card had not been a one-off by following up with the Radeon 2 (R200) core in 2001, marketed as the 8500. While again it lacked a little in raw power compared to nVidia's offerings, it offered great visual quality, strong DirectX 8.1 shader performance, and a rich feature set. As with other ATi products, it proved popular with OEMs, partly because it offered wider motherboard compatibility than nVidia's offerings.

Driver support continued to be an area of weakness for ATi, although over time, this was considerably improved. The 8500 established ATi as a serious performance and feature-integrated chipset competitor to nVidia.

The year 2002 saw ATi introducing a new Radeon architecture. Designed from the ground up for DirectX 9 operation, the 9700